

## KHON KAEN, Thailand

WMO: 483810

Lat: 16.467N

Lon: 102.784E

Elev: 670

StdP: 14.34

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | 58.4       | 61.0 | 44.8                          | 44.9 | 69.5 | 48.6 | 52.0 | 70.9 | 14.5                  | 75.5 | 13.3 | 77.1 | 4.6                   | 10   | 0.323 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 18.9                   | 100.8           | 76.2 | 98.7 | 76.2 | 96.7 | 76.5 | 81.6                | 91.2 | 80.8 | 90.1 | 80.1 | 89.0 | 6.0                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 79.2                            | 155.2 | 84.9 | 78.8 | 153.4 | 84.5 | 78.2 | 150.3 | 83.9 | 46.1          | 91.3 | 45.2 | 90.2 | 44.4 | 89.7 | 85.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 13.1              | 11.5 | 10.0 | DB                         | 52.8 | 103.6              | 3.5 | 2.0                                                | 50.3 | 105.0      | 48.2 | 106.2      | 46.2 | 107.3      | 43.7 | 108.7 |
|                   |      |      | WB                         | 49.5 | 83.5               | 3.2 | 1.3                                                | 47.2 | 84.5       | 45.3 | 85.2       | 43.5 | 86.0       | 41.2 | 86.9  |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 81.5  | 75.5  | 79.3  | 83.9  | 86.4  | 85.4  | 84.7  | 83.3  | 82.7  | 82.1  | 78.8  | 75.2  | (6)  |      |
| (7)  |                                                                                  | DBStd       | 5.11  | 4.69  | 5.35  | 5.10  | 3.99  | 3.17  | 2.47  | 2.46  | 2.17  | 2.04  | 2.40  | 3.65  | 4.73 | (7)  |
| (8)  |                                                                                  | HDD50       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD65       | 4     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2    | (9)  |
| (10) |                                                                                  | CDD50       | 11512 | 790   | 822   | 1049  | 1093  | 1096  | 1042  | 1033  | 1013  | 963   | 966   | 863   | 782  | (10) |
| (11) |                                                                                  | CDD65       | 6041  | 326   | 402   | 585   | 643   | 631   | 592   | 568   | 548   | 513   | 501   | 413   | 319  | (11) |
| (12) |                                                                                  | CDH74       | 65981 | 2939  | 4303  | 7220  | 8332  | 7576  | 7053  | 6261  | 5672  | 5008  | 4927  | 3893  | 2797 | (12) |
| (13) |                                                                                  | CDH80       | 29086 | 1130  | 2048  | 3878  | 4595  | 3668  | 3178  | 2481  | 2048  | 1702  | 1827  | 1528  | 1003 | (13) |
| (14) | Wind                                                                             | WSAvg       | 4.2   | 4.3   | 4.0   | 4.1   | 3.8   | 3.9   | 4.5   | 4.6   | 4.4   | 3.5   | 4.1   | 4.6   | 4.9  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 45.9  | 0.3   | 0.5   | 1.3   | 2.5   | 6.3   | 6.5   | 6.3   | 7.8   | 9.6   | 3.9   | 0.8   | 0.2  | (15) |
| (16) |                                                                                  | PrecMax     | 57.6  | 2.5   | 3.0   | 3.0   | 5.5   | 15.0  | 18.4  | 10.0  | 11.9  | 16.6  | 7.7   | 4.0   | 1.1  | (16) |
| (17) |                                                                                  | PrecMin     | 30.6  | 0.0   | 0.0   | 0.0   | 0.2   | 0.7   | 1.5   | 1.6   | 3.9   | 4.2   | 0.2   | 0.0   | 0.0  | (17) |
| (18) |                                                                                  | PrecStd     | 7.3   | 0.7   | 0.8   | 0.9   | 1.6   | 4.1   | 4.4   | 2.4   | 2.2   | 3.4   | 2.5   | 1.3   | 0.3  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 94.3  | 98.9  | 103.1 | 104.3 | 101.2 | 98.4  | 96.8  | 94.7  | 93.4  | 93.4  | 94.8  | 93.3 | (19) |
| (20) |                                                                                  |             | MCWB  | 71.8  | 72.8  | 73.3  | 76.8  | 79.1  | 78.5  | 78.5  | 79.4  | 79.5  | 78.1  | 75.7  | 73.8 | (20) |
| (21) |                                                                                  | 2%          | DB    | 91.5  | 96.7  | 100.3 | 101.9 | 98.5  | 95.4  | 93.5  | 92.1  | 91.4  | 91.3  | 91.7  | 90.3 | (21) |
| (22) |                                                                                  |             | MCWB  | 71.9  | 72.7  | 74.0  | 76.9  | 78.7  | 78.6  | 78.5  | 79.0  | 79.0  | 77.2  | 75.6  | 73.3 | (22) |
| (23) |                                                                                  | 5%          | DB    | 89.2  | 94.0  | 98.1  | 99.0  | 95.5  | 93.4  | 91.5  | 90.6  | 89.7  | 89.7  | 89.8  | 88.1 | (23) |
| (24) |                                                                                  |             | MCWB  | 71.4  | 73.1  | 74.4  | 77.3  | 78.7  | 78.7  | 78.6  | 78.7  | 78.3  | 76.4  | 74.8  | 72.2 | (24) |
| (25) |                                                                                  | 10%         | DB    | 86.3  | 91.2  | 95.3  | 96.6  | 93.2  | 91.6  | 89.7  | 88.9  | 88.0  | 88.2  | 87.9  | 85.8 | (25) |
| (26) |                                                                                  |             | MCWB  | 70.3  | 73.0  | 74.9  | 77.6  | 78.7  | 78.7  | 78.2  | 78.5  | 78.2  | 75.9  | 73.8  | 70.9 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 77.0  | 79.0  | 80.7  | 82.7  | 82.7  | 81.8  | 81.5  | 81.7  | 81.8  | 80.6  | 79.7  | 78.2 | (27) |
| (28) |                                                                                  |             | MCDWB | 87.3  | 90.6  | 94.2  | 95.9  | 93.3  | 90.6  | 89.6  | 89.5  | 88.5  | 87.9  | 89.0  | 87.1 | (28) |
| (29) |                                                                                  | 2%          | WB    | 75.1  | 77.2  | 79.1  | 80.9  | 81.2  | 80.8  | 80.3  | 80.5  | 80.5  | 79.4  | 78.1  | 75.6 | (29) |
| (30) |                                                                                  |             | MCDWB | 86.1  | 89.6  | 92.8  | 93.3  | 91.8  | 89.9  | 88.5  | 88.0  | 87.2  | 86.6  | 87.1  | 85.4 | (30) |
| (31) |                                                                                  | 5%          | WB    | 73.6  | 75.9  | 77.7  | 79.8  | 80.3  | 80.0  | 79.6  | 79.7  | 79.8  | 78.6  | 76.9  | 73.9 | (31) |
| (32) |                                                                                  |             | MCDWB | 84.4  | 87.7  | 90.9  | 92.2  | 90.6  | 88.9  | 87.5  | 86.9  | 86.1  | 85.6  | 85.4  | 84.8 | (32) |
| (33) |                                                                                  | 10%         | WB    | 72.2  | 74.6  | 76.6  | 78.8  | 79.6  | 79.4  | 78.9  | 79.1  | 79.0  | 77.9  | 75.8  | 72.5 | (33) |
| (34) |                                                                                  |             | MCDWB | 82.8  | 86.1  | 89.1  | 91.3  | 89.2  | 87.9  | 86.5  | 85.9  | 85.1  | 84.4  | 84.0  | 83.3 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 21.8  | 21.8  | 20.5  | 18.9  | 15.9  | 14.4  | 13.1  | 12.8  | 12.7  | 14.7  | 18.2  | 20.6 | (35) |
| (36) |                                                                                  |             | MCDBR | 23.5  | 24.4  | 23.9  | 22.2  | 19.1  | 16.9  | 15.9  | 15.4  | 14.9  | 16.4  | 19.5  | 21.6 | (36) |
| (37) |                                                                                  |             | MCWBR | 8.6   | 7.6   | 7.0   | 6.3   | 5.8   | 5.1   | 5.1   | 5.3   | 5.2   | 5.2   | 6.8   | 8.0  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 20.5  | 20.4  | 20.6  | 19.6  | 16.9  | 15.0  | 14.1  | 13.9  | 13.5  | 14.2  | 17.5  | 20.3 | (38) |
| (39) |                                                                                  |             | MCWBR | 8.4   | 7.5   | 7.6   | 6.9   | 6.0   | 5.5   | 5.4   | 5.6   | 5.5   | 5.4   | 7.0   | 8.6  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.475 | 0.582 | 0.651 | 0.612 | 0.506 | 0.458 | 0.446 | 0.456 | 0.500 | 0.536 | 0.473 | 0.468 | (40) |      |
| (41) |                                                                                  | taud        | 2.087 | 1.822 | 1.702 | 1.838 | 2.161 | 2.317 | 2.348 | 2.307 | 2.172 | 2.032 | 2.181 | 2.130 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 256   | 234   | 223   | 233   | 255   | 265   | 268   | 268   | 257   | 244   | 256   | 254   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 49    | 67    | 78    | 68    | 49    | 41    | 40    | 42    | 48    | 54    | 45    | 46    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1543  | 1643  | 1677  | 1809  | 1741  | 1633  | 1520  | 1485  | 1459  | 1558  | 1552  | 1497  | (44) |      |
| (45) |                                                                                  | RadStd      | 94    | 82    | 85    | 74    | 91    | 86    | 105   | 88    | 98    | 114   | 57    | 39    | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (21 neighbors) |  | +0.54 | N/A     | N/A    | N/A     | +0.31 | +0.25 | N/A         | N/A   | +216  | +220  |

Nomenclature: See separate page